

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO3: Analyze knowledge representation and reasoning using propositional logic, FOL, inference, resolution, chaining methods, knowledge representation to perform logical reasoning for drawing valid conclusions and decision making. (BL4)

Assessment Tool Weightage: 50%

Time Duration: 1 Hour

QUESTIONS: 4

Total Marks: 40

Annexure A

CASE STUDY

Code of Conduct:

I S. Chandra Shekhar Reddy / 192372163 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

SIMATS ENGINEERING

ASSESSMENT TOOL 1 — ANSWER SCRIPT

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Register No: 192372163

Course: Artificial Intelligence (CSA17)

Course Outcome: CO3 — Knowledge representation and reasoning using propositional logic, FOL, inference, resolution, chaining methods (BL4)

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Case Study 1 — Smart Medical Diagnosis System

(a) Propositional Logic Representation

Symbol definitions:

- F = Patient has Fever
- C = Patient has Cough
- R = Patient has Rash
- B = Patient has Breathlessness
- Flu = Possible Flu
- Measles = Possible Measles
- Pneumonia = Possible Pneumonia

Rules in Propositional Logic:

Rule 1: $F \wedge C \rightarrow \text{Flu}$

Rule 2: $F \wedge R \rightarrow \text{Measles}$

Rule 3: $C \wedge B \rightarrow \text{Pneumonia}$

Facts for Patient A:

F = True, C = True, B = True

(b) Forward Chaining

Known facts: F = True, C = True, B = True

Step 1: Test Rule 1 ($F \wedge C \rightarrow \text{Flu}$). F=T, C=T $\rightarrow$ condition satisfied $\rightarrow$ FIRE $\rightarrow$ derive Flu = True

Step 2: Test Rule 2 ($F \wedge R \rightarrow \text{Measles}$). R is not asserted (unknown) $\rightarrow$ condition NOT satisfied $\rightarrow$ cannot fire

Step 3: Test Rule 3 ($C \wedge B \rightarrow \text{Pneumonia}$). C=T, B=T $\rightarrow$ condition satisfied $\rightarrow$ FIRE $\rightarrow$ derive Pneumonia = True

Step 4: No further rules can fire (all antecedents already used).

Derived diagnoses for Patient A: Flu, Pneumonia (Measles is not derived since Rash is not known to be true).

(c) Backward Chaining — Goal: 'Patient A has Pneumonia'

Goal: Pneumonia

Pneumonia is the consequent of Rule 3: $C \wedge B \rightarrow \text{Pneumonia}$

Sub-goal 1: C (Cough) $\rightarrow$ matches known fact C = True $\rightarrow$ satisfied

Sub-goal 2: B (Breathlessness) $\rightarrow$ matches known fact B = True $\rightarrow$ satisfied

Both sub-goals proven $\rightarrow$ Goal 'Pneumonia' is PROVED True.

Goal reduction tree:

Pneumonia

- ├ C (Cough) — fact, True
- └ B (Breathlessness) — fact, True

Case Study 2 — Autonomous Traffic Management Agent

(a) Unification of Rule 1 with the Facts

Rule 1: $\forall x: \text{Vehicle}(x) \wedge \text{EmergencyType}(x) \rightarrow \text{ClearPath}(x)$

Facts: $\text{Vehicle}(\text{Ambulance}), \text{EmergencyType}(\text{Ambulance})$

Step 1: $\text{UNIFY}(\text{Vehicle}(x), \text{Vehicle}(\text{Ambulance})) \rightarrow \theta_1 = \{x/\text{Ambulance}\}$

Step 2: Apply θ_1 to $\text{EmergencyType}(x) \rightarrow \text{EmergencyType}(\text{Ambulance})$; UNIFY with fact $\text{EmergencyType}(\text{Ambulance}) \rightarrow \theta_2 = \{x/\text{Ambulance}\}$ (consistent with θ_1)

MGU: $\theta = \{x/\text{Ambulance}\}$

Applying θ to the consequent gives: $\text{ClearPath}(\text{Ambulance})$

(b) Resolution Proof that $\text{Proceed}(\text{CarA})$ is True

Step 1 — Convert rules to CNF (implication elimination: $A \rightarrow B \equiv \neg A \vee B$):

C1: $\neg \text{Vehicle}(x) \vee \neg \text{EmergencyType}(x) \vee \text{ClearPath}(x)$ [Rule 1]

C2: $\neg \text{ClearPath}(x) \vee \text{GreenSignal}(x)$ [Rule 2, GreenSignal part]

C3: $\neg \text{Vehicle}(x) \vee \neg \text{Behind}(x, y) \vee \neg \text{GreenSignal}(y) \vee \text{Proceed}(x)$ [Rule 3]

Facts (unit clauses):

C4: $\text{Vehicle}(\text{Ambulance})$ C5: $\text{EmergencyType}(\text{Ambulance})$

C6: $\text{Vehicle}(\text{CarA})$ C7: $\text{Behind}(\text{CarA}, \text{Ambulance})$

Negated goal:

C8: $\neg \text{Proceed}(\text{CarA})$

Resolution steps:

Step 1: Resolve C1{x/Ambulance} with C4 $\rightarrow \neg \text{EmergencyType}(\text{Ambulance}) \vee \text{ClearPath}(\text{Ambulance})$ [C9]

Step 2: Resolve C9 with C5 $\rightarrow \text{ClearPath}(\text{Ambulance})$ [C10]

Step 3: Resolve C2{x/Ambulance} with C10 $\rightarrow \text{GreenSignal}(\text{Ambulance})$ [C11]

Step 4: Resolve C3{x/CarA, y/Ambulance} with C6 $\rightarrow \neg \text{Behind}(\text{CarA}, \text{Ambulance}) \vee \neg \text{GreenSignal}(\text{Ambulance}) \vee \text{Proceed}(\text{CarA})$ [C12]

Step 5: Resolve C12 with C7 $\rightarrow \neg \text{GreenSignal}(\text{Ambulance}) \vee \text{Proceed}(\text{CarA})$ [C13]

Step 6: Resolve C13 with C11 $\rightarrow \text{Proceed}(\text{CarA})$ [C14]

Step 7: Resolve C14 with negated goal C8 ($\neg \text{Proceed}(\text{CarA})$) $\rightarrow$ empty clause $\square$

Since resolution derives the empty clause, $\text{Proceed}(\text{CarA})$ is PROVED true.

(c) Sufficiency of the KB for Two Simultaneous Emergency Vehicles

The current KB is NOT sufficient. It has no predicate or rule to handle conflict between two vehicles that both satisfy $\text{Vehicle}(x) \wedge \text{EmergencyType}(x)$ at the same junction — as written, Rule 1 and Rule 2 would independently grant ClearPath and GreenSignal to both vehicles, which is logically derivable but physically unsafe (two ambulances cannot both get the green light on conflicting approaches).

Additional rules/facts needed:

- A conflict predicate, e.g. $\text{ConflictingApproach}(x, y)$, to represent that two vehicles' paths intersect.
- A priority predicate/fact, e.g. $\text{Priority}(x, n)$ based on emergency severity, distance, or arrival time.
- A tie-breaking rule: $\forall x \forall y: \text{Vehicle}(x) \wedge \text{Vehicle}(y) \wedge \text{EmergencyType}(x) \wedge \text{EmergencyType}(y) \wedge \text{ConflictingApproach}(x, y) \wedge \text{HigherPriority}(x, y) \rightarrow \text{ClearPath}(x) \wedge \neg \text{ClearPath}(y)$.

- A mutual-exclusion constraint ensuring GreenSignal cannot hold for two conflicting approaches at once.

Without these additions, the KB is logically incomplete for the real-world safety requirement, even though it remains internally consistent as a set of Horn clauses.

Case Study 3 — Agricultural Expert Reasoning System

(a) CNF Conversion of P1, P2, P3 and Facts

P1: SoilDry $\rightarrow$ IrrigationNeeded

Eliminate implication: $\neg$ SoilDry $\vee$ IrrigationNeeded (already in CNF)
 $\rightarrow$ C1

P2: IrrigationNeeded $\wedge$ CropWheat $\rightarrow$ ApplyDripMethod

Eliminate implication: $\neg$ (IrrigationNeeded $\wedge$ CropWheat) $\vee$ ApplyDripMethod

Move $\neg$ inward (De Morgan): $\neg$ IrrigationNeeded $\vee$ $\neg$ CropWheat $\vee$ ApplyDripMethod
 $\rightarrow$ C2 (single clause, no distribution needed)

P3: $\neg$ ApplyDripMethod $\wedge$ CropWheat $\rightarrow$ CropAtRisk

Eliminate implication: $\neg$ ($\neg$ ApplyDripMethod $\wedge$ CropWheat) $\vee$ CropAtRisk

Move $\neg$ inward: ApplyDripMethod $\vee$ $\neg$ CropWheat $\vee$ CropAtRisk $\rightarrow$ C3

Facts (already unit clauses):

C4: SoilDry

C5: CropWheat

(b) Resolution Refutation — Prove 'ApplyDripMethod'

Negate goal: C6: $\neg$ ApplyDripMethod

Step 1: Resolve C1 ($\neg$ SoilDry $\vee$ IrrigationNeeded) with C4 (SoilDry) $\rightarrow$
 IrrigationNeeded [C7]

Step 2: Resolve C2 ($\neg$ IrrigationNeeded $\vee$ $\neg$ CropWheat $\vee$ ApplyDripMethod) with C7
 $\rightarrow$ $\neg$ CropWheat $\vee$ ApplyDripMethod [C8]

Step 3: Resolve C8 with C5 (CropWheat) $\rightarrow$ ApplyDripMethod [C9]

Step 4: Resolve C9 with C6 ($\neg$ ApplyDripMethod) $\rightarrow$ empty clause $\square$

The derivation of $\square$ proves ApplyDripMethod by refutation.

Resolution proof tree:

C1 + C4 $\rightarrow$ C7 (IrrigationNeeded)

C7 + C2 $\rightarrow$ C8 ($\neg$ CropWheat $\vee$ ApplyDripMethod)

C8 + C5 $\rightarrow$ C9 (ApplyDripMethod)

C9 + C6 ($\neg$ ApplyDripMethod) $\rightarrow$ $\square$

(c) Effect of Removing the Fact 'SoilDry'

Without C4 (SoilDry), Step 1 above can no longer be performed — C1 ($\neg$ SoilDry $\vee$ IrrigationNeeded) has no matching unit clause to resolve against, so IrrigationNeeded can no longer be derived. Consequently C2 cannot fire, and ApplyDripMethod can no longer be derived either.

Whether 'CropAtRisk' still follows now depends on the reasoning assumption used:

- Under the Closed World Assumption (CWA) / negation-as-failure, commonly used by rule-based expert systems: since ApplyDripMethod cannot be proven, it is treated as False, so $\neg$ ApplyDripMethod is True; combined with CropWheat = True, rule P3 fires and CropAtRisk IS derived.

- Under strict logical entailment (Open World Assumption, standard FOL semantics): ApplyDripMethod is merely unknown, not proven false, so $\neg \text{ApplyDripMethod}$ cannot be asserted True — CropAtRisk does NOT follow.

Conclusion: 'CropAtRisk' follows only if the system reasons under the Closed World Assumption; under pure logical entailment it remains unproven.

Case Study 4 — Wumpus World Knowledge Agent

(a) Belief State from Percepts — Modus Ponens

Percepts: Stench[1,2], Breeze[1,1], Glitter[2,2]

Modus Ponens 1: Stench[1,2] is True, Rule 1: Stench[1,2] $\rightarrow$ Wumpus adjacent to [1,2] $\Rightarrow$ derive: Wumpus is adjacent to [1,2]

Modus Ponens 2: Breeze[1,1] is True, Rule 2: Breeze[1,1] $\rightarrow$ Pit adjacent to [1,1] $\Rightarrow$ derive: Pit is adjacent to [1,1]

Modus Ponens 3: Glitter[2,2] is True, Rule 3: Glitter[x,y] $\rightarrow$ Gold at [x,y] $\Rightarrow$ derive: Gold is at [2,2]

Derived conclusions:

- Wumpus is in one of the cells adjacent to [1,2] $\rightarrow$ candidates {[1,1], [2,2], [1,3]}
- Pit is in one of the cells adjacent to [1,1] $\rightarrow$ candidates {[1,2], [2,1]}
- Gold is located at [2,2]

(b) Forward Chaining to Locate Wumpus and Pit

Step 1: Breeze[1,1] fires Rule 2 $\rightarrow$ Pit $\in$ {[1,2], [2,1]} (cells adjacent to [1,1])

Step 2: Stench[1,2] fires Rule 1 $\rightarrow$ Wumpus $\in$ {[1,1], [2,2], [1,3]} (cells adjacent to [1,2])

Step 3: [1,1] is the agent's starting cell and the agent survives there, so by convention Wumpus $\neq$ [1,1] and Pit $\neq$ [1,1] $\rightarrow$ eliminate [1,1] from the Wumpus candidate set

Step 4: Refined Wumpus location: {[2,2], [1,3]}

Step 5: Refined Pit location: {[1,2], [2,1]} (no further percept available yet to eliminate either)

Forward chaining narrows the Wumpus to two candidate cells and the Pit to two candidate cells; further percepts (e.g., Breeze or Stench at [2,1] or [1,3]) are required to pinpoint exact locations.

(c) Is the Path [1,1] $\rightarrow$ [1,2] $\rightarrow$ [2,2] Safe?

Applying Rule 4 ($\neg \text{Wumpus}[x,y] \wedge \neg \text{Pit}[x,y] \rightarrow \text{Safe}[x,y]$) to each cell on the path:

- [1,2]: Stench was perceived here, meaning the Wumpus is adjacent to [1,2], not necessarily inside it — so [1,2] is not proven to contain the Wumpus. However, [1,2] is itself one of the two candidate cells for the Pit (from part a/b). Since $\neg \text{Pit}[1,2]$ cannot yet be proven, Safe[1,2] cannot be established by Rule 4.
- [2,2]: This cell holds the Glitter/Gold, but it is also one of the two candidate cells for the Wumpus (from part b). Since $\neg \text{Wumpus}[2,2]$ cannot yet be proven, Safe[2,2] cannot be established by Rule 4 either.

Conclusion: The path [1,1] $\rightarrow$ [1,2] $\rightarrow$ [2,2] CANNOT be certified as safe with the current knowledge — both [1,2] and [2,2] remain live candidates for a Pit and the Wumpus respectively. The agent should not take this path yet. It should first gather more percepts (e.g., visit [2,1] to check for Breeze, or otherwise probe [1,3]) to eliminate one of the candidates in each set before Rule 4 can certify Safe[1,2] and Safe[2,2].